

Xiang Liu

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EDUCATION

Stony Brook University

B.S. in Computer Science Honors

Stony Brook, NY

Expected May 2027

- **GPA:** 3.95/4.00 | Dean's List (6 Semesters)
- **Coursework:** Data Structures & Algorithms, Cloud Computing, Systems, Machine Learning, Software Eng.

TECHNICAL SKILLS

Programming: Python, Java, C, TypeScript, JavaScript, MongoDB, R, OCaml, HTML/CSS, Git, Bash, Claude Code

Tools: PyTorch, AWS, Snowflake, SQL, MongoDB, Databricks, Supabase, FastAPI, Scikit-learn, RESTful API, Next.js, React

EXPERIENCE

Capital One

June 2026 – August 2026

Software Engineering Intern; Card Tech Data Team

Richmond, VA

- Architected a dataset quality scoring system from scratch in Python as part of a 8-engineer team, scanning 2,000+ GitHub repositories via static code analysis to validate data lineage and dataset trustworthiness
- Co-developed a graph algorithm to construct end-to-end data lineage graphs consisting of 5,000+ financial datasets, enabling change-impact analysis that exposed data movements undetected by existing tools
- Deployed and provisioned the production CI/CD pipeline across Databricks, AWS DynamoDB and Aurora RDS
- Delivered a full-stack Typescript UI surfacing real-time dataset quality scores, translating complex lineage data into an accessible view for Risk Managers and Analysts
- **Impact:** Saved \$300k in costs tied to manual lineage reviews, improving data observability across the entire lineage

Learning & Intelligence Lab

May 2025 – August 2025

Research Intern; advised by Professor Jorge-Mendez Mendez

Stony Brook, NY

- Applied image-generation diffusion and flow-matching models to generate executable action parameters for home robots, adapting image-generation techniques to a robotics planning domain
- Optimized the sampling pipeline to cut inference time 30% while preserving action-sampling accuracy
- Partnered with 2 graduate researchers to design evaluation methods, benchmarks measuring model bias and reliability

SUNY Research Foundation

May 2024 – August 2024

AI4Science REU Research Intern; advised by Professor Yi Liu

Stony Brook, NY

- Built deep learning models including Fourier Neural Operators, U-Net, and CNNs for weather forecasting
- Delivered a 24-hour-ahead temperature, precipitation forecasting model in the US, validated against observed data
- Engineered an ETL pipeline that mined 100GB+ of meteorological data, automating preparation for training
- **Impact:** Compressed inference from 3+ hours to minutes, making prediction accessible without HPC resources

PROJECTS

Course Advising Bot | *LangChain, TypeScript, FastAPI, Pinecone, vLLM, Supabase, SQL*

January 2026 – August 2026

- Developed a full-stack AI course advising chatbot to deliver real-time, accurate academic guidance, serving 400+ Stony Brook students and reducing reliance on limited advisor availability
- Engineered a NLP Retrieval-Augmented Generation (RAG) information retrieval pipeline integrating vector embeddings with self-hosted vLLM models, reducing hallucinated answers and improving factual accuracy by 80%
- Customized Python web scraper to extract 200+ pages of undergrad bulletin data, enabling source-grounded answers
- Architected a thread-based memory management system with LangGraph, PostgreSQL and Supabase, allowing users to create, store and resume multiple concurrent conversation states that are context-aware
- Built a responsive, production-ready frontend with Next.js and FastAPI backend, featuring streaming responses and a chat interface modeled on modern LLM products

Car Brand Detector | *Python, PyTorch, Scikit-Learn*

August 2025 – December 2025

- Engineered an end-to-end image classification pipeline in PyTorch, training CNN, VGG-16, and ResNet18 models on an 8,000+ image dataset to accurately identify car brands
- Leveraged transfer learning to fine-tune ResNet18 model, boosting validation accuracy by 10% and reduce overfitting
- Enhanced system transparency and user trust by integrating GradCAM, visualizing model decision making instead of being a "black-box" model

LEADERSHIP ACTIVITIES

Stony Brook University Campus Residence

August 2024 – Present

Resident Assistant

Stony Brook, NY

- Led 8 campus-wide events with over 600 attendees collaborating with 10+ RAs, 650% more than average
- Served as first responder for 6+ residential emergencies, coordinating with supervisors, campus police to stabilize incidents
- Mentored 40 residents consisted of first-year, international and transfer students to foster community building
- **Impact:** 95% retention rate of residents returning next year and gained \$1,000 extra funding for future events